CLINIC DASHBOARD

My Data Cleaning Process

Messy raw patient appointment data → clean, analysis-ready data

I worked with a patient appointment dataset containing 1,000 data rows. I mainly used Excel for this exercise because I wanted the cleaning process to be simple, visible, and repeatable.

1. What the raw data looked like

The raw data looked usable at first, but it had quite a few inconsistencies. The main columns were patient ID, patient name, age, gender, appointment date, booking date, doctor, department, billing amount and follow-up.

Gender was entered in different ways: M, Male, male, F, Female, female, 1 and 0. Follow-up had a similar problem: Y, Yes, N, No, 1 and 0.

Dates were mixed too. I found examples such as 26/02/2026, 05/23/2025, 30-Nov-25 and 18-May-25 in the same dataset. Billing amounts were also inconsistent, for example $84.44, Rs85.76, Â£425.8 and â‚¬344.26. There were blank values, especially in fields such as billing amount and gender.

I also noticed that patient IDs repeat. I did not automatically call those duplicates because a repeated ID does not necessarily mean the whole record is duplicated.

2. Tools and techniques I used

For this particular cleaning exercise I used Excel rather than Python/pandas, SQL or OpenRefine. My main tools were Excel Tables, filters and formulas.

- Ctrl + T to turn the data into an Excel Table.
- TRIM, CLEAN and PROPER for text cleaning.
- LET, IF and IFERROR for standardizing values and handling missing/problem values.
- VALUE and SUBSTITUTE for converting billing text into numbers.
- Filters and Number Filters for finding missing and invalid values.
- COUNTIFS / duplicate checking for reviewing repeated records.

3. The cleaning process I followed

Step 1 – I kept the raw data safe

I left the original data in a Raw_Data sheet and created a separate Cleaned_Data sheet. I prefer this approach because I can always compare my cleaned value with the original.

Step 2 – I cleaned patient names

I used =PROPER(TRIM(CLEAN(B2))). TRIM removed unnecessary spaces, CLEAN removed unwanted characters, and PROPER made the capitalization consistent.

Step 3 – I standardized gender

I converted F, Female and 1 to Female, and M, Male and 0 to Male. Blank values were marked Missing and unexpected values were marked Review.

Step 4 – I cleaned the dates

I converted recognizable date values into Excel dates and used a consistent display format. I was careful with dates such as 05/11/2025 because that can mean different things depending on the original date convention. I did not want to guess and create a wrong date, so ambiguous cases were left for review.

STEP 5 – I CLEANED DOCTOR AND DEPARTMENT

I used TRIM, CLEAN and PROPER to make the text consistent. I did not automatically remove titles such as MD, DDS or DVM because that would be a separate business decision.

Step 6 – I cleaned billing amount

I removed currency symbols such as $, Rs, £ and €. Then I converted the remaining text to a real Excel number. Blank billing values were labelled Missing, and values that could not be converted were labelled Review.

Step 7 – I standardized follow-up

Y, Yes and 1 became Yes. N, No and 0 became No. Blank values were marked Missing.

Step 8 – I checked age

I checked for blank ages, text values, values below 0 and values above 120. I used an Age_Status column to label records Valid, Missing, Text or Invalid.

Step 9 – I checked missing values and duplicates

I filtered the cleaned data to find records needing attention. I checked complete records before treating them as duplicates instead of removing rows based only on patient ID.

Step 10 – I added a final quality flag

I added a simple OK / Needs Review status. This made it easy to filter the dataset and see what still needed a human check.

4. Before and after examples

Field	Raw value	Clean value
Gender	F	Female
Gender	1	Female
Gender	0	Male
Follow-up	Y	Yes
Follow-up	0	No
Billing	Rs85.76	85.76
Billing	â‚¬203.34	203.34
Billing	Â£425.8	425.80
Billing	(blank)	0.00
Name	tammy williams	Tammy Williams

Power BI Clinic Dashboard

The dashboard was created using the ClinicMessyDataSet.xlsx file. The main purpose was to show clinic billing and patient information in a simple way that can be used during a review presentation. The dashboard allows the user to compare billing between departments and filter the results using a department slicer.

1. Open Power BI and import the Excel file

Opened Power BI Desktop and selected Home → Get data → Excel workbook. Selected ClinicMessyDataSet.xlsx and opened it. In the Navigator window, the available sheets were displayed.

2. Select the Clean_Data sheet

Selected the Clean_Data sheet from the Navigator. This sheet was used instead of the raw data because it contains the cleaned and organized fields needed for the report. Clicked Transform Data to check the data before loading it.

3. Check and clean the data in Power Query

In Power Query Editor, checked the important columns. Doctor_Department was kept as text, Billing_amt was checked as a numeric/decimal field, and the date columns were checked as Date. Also checked for obvious errors or incorrect values before loading the data.

4. Load the data into Power BI

After checking the data, selected Home → Close & Apply. Power BI then loaded the Clean_Data table into the report and data model.

5. Create the Total Billing measure

Created a new measure from the Clean_Data table. The measure calculates the total of the Billing_amt column. This measure is used in the main chart and the Total Billing KPI card.

6. Create the Patient Count measure

Created a Patient Count measure using the Patient_id field. DISTINCTCOUNT was used so the report can count unique patients rather than simply counting rows.

7. Create the Average Billing measure

Created an Average Billing measure using the Billing_amt column. This gives the average billing amount and is used in the KPI card and the detailed table.

8. Create the main column chart

Moved to Report view and selected a blank area of the report canvas. From the Visualizations pane, selected Clustered column chart. Added Doctor_Department to the X-axis and Total Billing to the Y-axis. This created a comparison of billing across departments.

9. Sort the main chart

Selected the three-dot menu on the chart and sorted the chart by Total Billing in descending order. This places the department with the highest billing at the top of the comparison and makes the chart easier to read.

10. Turn on data labels

Selected the chart and opened Format visual. Under Data labels, turned the labels on. The display units and decimal places were adjusted so the billing values are easy to read directly on the columns.

11. Add the KPI cards

Added three Card visuals to the top of the report. The first card uses Total Billing, the second uses Patient Count, and the third uses Average Billing. These cards provide a quick summary before the reviewer looks at the detailed chart.

12. Add the department slicer

Added a Slicer visual and placed Doctor_Department into it. The slicer allows the user to select a department and automatically filter the other visuals on the page.

13. Change the slicer to Dropdown

Selected the Doctor_Department slicer and opened Format visual → Visual → Slicer settings. Changed the slicer style to Dropdown. This keeps the dashboard compact and avoids taking too much space on the page.

14. Enable slicer search

Search was enabled for the department slicer where available. This makes it easier to find a department when the list becomes longer.

15. Add chart tooltips

Selected the main chart and added Patient Count and Average Billing to the Tooltips field. When the user hovers over a department column, Power BI can show additional information without adding more text to the chart itself.

16. Create the summary table

Added a Table visual below the main chart. Added Doctor_Department, Total Billing, Patient Count and Average Billing. The table provides the exact values behind the visual comparison.

17. Add conditional formatting

Selected the Total Billing field in the table and used Conditional formatting → Background color. A color scale was applied so higher and lower billing values are easier to identify at a glance.

18. Arrange the dashboard

The three KPI cards were placed across the top. The department slicer was placed near the top for easy access. The main billing chart was kept large in the middle, and the detailed table was placed below it.

19. Check the dashboard interaction

Tested the department slicer by selecting individual departments such as Cardiology. Checked that the KPI cards, chart and table responded to the selection. Then cleared the selection and confirmed that the full dashboard values returned.

20. Final formatting and review

Checked the chart title, card titles, data labels, spacing and overall readability. Kept the design simple and avoided too many colors or unnecessary visuals. The final dashboard was prepared so it can be used during a review presentation.